

Maximum and minimum values



- 1 Consider the function $f(x) = 2x^3 - 12x^2 + 18x + 3$.
 - a Find the x -coordinates of the turning points.
 - b State the coordinates of the local maximum.
 - c State the coordinates of the local minimum.
 - d State the absolute maximum value on $[0, 7]$.
 - e State the absolute minimum value on $[0, 7]$.

- 2 Consider the function $y = 5x^3 + 7x^2 + 3x + 4$ with restricted domain of $[-1, 1]$.
 - a Find the y -intercept.
 - b Find the stationary points.
 - c Determine the nature of the stationary points.
 - d Find any points of inflection and classify them.
 - e Describe the behaviour of the function as $x \rightarrow \pm\infty$.
 - f Find the maximum and minimum value of the function for the restricted domain.

- 3 Consider the function $y = (x^2 - 1)^2 + 4$ with restricted domain of $[-2, 2]$.
 - a Find the y -intercept.
 - b Find the x -intercept(s).
 - c Determine whether the function is odd or even.
 - d Find the stationary points.
 - e Determine the nature of the stationary points.
 - f Find any points of inflection and classify them.
 - g Describe the behaviour of the function as $x \rightarrow \pm\infty$.
 - h Find the maximum and minimum value of the function for the restricted domain.

- 4 Consider the function $y = 3x^4 + 4x^3 - 12x^2 - 1$.
 - a Find the y -intercept.
 - b Find the stationary points.
 - c Determine the nature of the stationary points.
 - d Find any points of inflection and classify them.
 - e Describe the behaviour of the function as $x \rightarrow \pm\infty$.
 - f Find the maximum and minimum value of the function for the restricted domain $[-1, 1]$.



Graphs of functions

- 5** Consider the function $y = -x^2 + 8x - 21$ with restricted domain of $[0, 6]$.
- a** Find the y -intercept.
 - b** Find the stationary points.
 - c** Determine the nature of the stationary points.
 - d** Find the maximum and minimum value of the function for the restricted domain.
 - e** Sketch the function on a number plane.
- 6** Consider the function $y = x^4 - 2x^2 + 1$ with restricted domain of $[-2, 3]$.
- a** Find the y -intercept.
 - b** Find the x -intercept(s).
 - c** Determine whether the function is odd or even.
 - d** Find the stationary points.
 - e** Determine the nature of the stationary points.
 - f** Find any points of inflection and classify them.
 - g** Find the maximum and minimum value of the function for the restricted domain.
 - h** Sketch the function on a number plane.
- 7** Consider the function $y = (2x - 1)^2(1 - x)$ with restricted domain of $[0, 2]$.
- a** Find the y -intercept.
 - b** Find the x -intercept(s).
 - c** Find the stationary points.
 - d** Determine the nature of the stationary points.
 - e** Find any points of inflection and classify them.
 - f** Describe the behaviour of the function as $x \rightarrow \pm\infty$.
 - g** Find the maximum and minimum value of the function for the restricted domain.
 - h** Sketch the function on a number plane.
- 8** Consider the function $y = 9x^2 + 18x - 16$.
- a** Find the y -intercept.
 - b** Find the x -intercept(s).
 - c** Find the stationary points.
 - d** Determine the nature of the stationary points.



9 Consider the function $y = 8(x - 1)^2(x + 2)$.

- a** Find the stationary points.
- b** Determine the nature of the stationary points.
- c** Determine the point of inflection.
- d** Sketch the function on a number plane.

10 For each of the following functions:

- i** Find any stationary points and determine their nature.
- ii** Determine the points of inflection.
- iii** Sketch the graph showing stationary points and points of inflection.
- iv** Write the domain for which the function is decreasing.

a $f(x) = 8 - x^3$

b $f(x) = 8 - x - x^3$

c $f(x) = x^3 - 3x^2 - 9x + 11$

d $y = x^3 - 6x^2 - 3$

e $f(x) = (x^2 - 9)^2 + 3$

f $f(x) = x^3 - 5x^2 + 3x - 5$

11 For each of the following functions:

- i** Find any stationary points and determine their nature.
- ii** Determine the points of inflection.
- iii** Sketch the graph showing stationary points and points of inflection.
- iv** Write the domain for which $\frac{dy}{dx}$ is positive.

a $f(x) = x^4(3x - 10)$

b $f(x) = x^2(x + 3)$

12 For each of the following functions:

- i** Find any stationary points and determine their nature.
- ii** Determine the points of inflection.
- iii** Sketch the graph showing stationary points and points of inflection.
- iv** Write the domain over which the curve is concave up and concave down.

a $f(x) = 3x^2 - x^3 - 4$

b $f(x) = x^3 - 3x^2 - 9x + 11$

c $f(x) = x^3(x - 4)$

d $f(x) = x^2(x^2 - 6x + 12)$

13 Consider the function $y = 2(x + 5)^3 + 3$.



- a Find the y -intercept.
- b Find the x -intercept(s).
- c Find the stationary points.
- d Determine the nature of the stationary points.
- e Find any points of inflection and classify them.
- f Describe the behaviour of the function as $x \rightarrow \pm\infty$.
- g Sketch the function on a number plane.

14 Consider the function $y = 2 + 9x - 3x^2 - x^3$.

- a Find the y -intercept.
- b Find the stationary points.
- c Determine the nature of the stationary points.
- d Find any points of inflection and classify them.
- e Describe the behaviour of the function as $x \rightarrow \pm\infty$.
- f Sketch the function on a number plane.

15 Consider the function $y = -8(x - 1)^2(x + 2)$.

- a Find the y -intercept.
- b Find the x -intercept(s).
- c Find the stationary points.
- d Determine the nature of the stationary points.
- e Find any points of inflection and classify them.
- f Describe the behaviour of the function as $x \rightarrow \pm\infty$.
- g Sketch the function on a number plane.

16 Consider the function $y = (2x + 1)(x + 2)^4$.

- a Find the y -intercept.
- b Find the x -intercept(s).
- c Find the stationary points.
- d Determine the nature of the stationary points.
- e Describe the behaviour of the function as $x \rightarrow \pm\infty$.
- f Sketch the function on a number plane.

17 Consider the function $y = \frac{8}{x^2 + 4}$.



- a Find the y -intercept.
- b Find the x -intercept(s).
- c Determine whether the function is odd or even.
- d Find the stationary points.
- e Determine the nature of the stationary points.
- f Describe the behaviour of the function as $x \rightarrow \pm\infty$.
- g Sketch the function on a number plane.